

SIMATS ENGINEERING
(SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – RESEARCH PAPER REVIEW / LITERATURE SURVEY

AI-Powered Smart Volunteer and NGO Resource Management Platform

(Assigned Problem Statement 28)

Course Code	CSA10	Course Title	Software Engineering
CO Assessed	CO5	Assessment Tool	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage	20%	Total Marks	25
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CO5 Statement: *Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly.*

Code of Conduct

I R. Indhu(192524332) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R. Indhu

1. Introduction

Non-Governmental Organisations (NGOs) and volunteer groups play a central role in delivering humanitarian aid, education, healthcare, disaster relief, and community-development programmes. As these organisations grow in scale and geographic reach, the manual coordination of volunteers, donations, and project activities becomes increasingly fragmented, causing delays, duplicated effort, and inefficient use of scarce resources. This literature survey examines the assigned problem statement — the design of an AI-Powered Smart Volunteer and NGO Resource Management Platform — which aims to digitally manage volunteer registrations, apply artificial intelligence to match volunteers with suitable activities, monitor project progress, track donations and resources, and provide real-time dashboards for administrators. The overarching goal of such a system is to improve transparency, volunteer engagement, and operational efficiency across the NGO sector.

1.1 Scope of the Survey

To understand the state of the art relevant to this problem, this survey reviews research spanning four closely related threads of prior work: (i) AI- and preference-based volunteer-to-task matching, (ii) digital NGO and volunteer management platforms, (iii) blockchain-based donation and fund-tracking systems, and (iv) GIS and optimisation-based resource allocation in humanitarian and disaster-relief logistics. Together, these threads cover the major functional pillars named in the problem statement — volunteer management, AI-based matching, donation tracking, resource optimisation, and project monitoring.

2. Objectives

- Manage volunteer registrations digitally.
- Match volunteers to activities using AI-based techniques.
- Monitor NGO project progress in real time.
- Track donations and resources transparently.
- Optimise volunteer and resource allocation.
- Generate organisational and impact reports.

- Improve stakeholder (donor, volunteer, beneficiary) communication.
- Enhance operational transparency and accountability.

3. Literature Review

2.1 Mobile Volunteer-Matching Systems

Chen, Cheng, Sandnes & Lee (2011)

Chen et al. designed a mobile matching prototype that helps non-profit organisations advertise ongoing volunteering opportunities on an interactive map, displaying opportunities as tappable icons so potential volunteers can browse activities near them. The prototype was implemented as an Android application and evaluated qualitatively through interviews with two organisation managers and three volunteers. The study found that a location-aware, visual browsing interface made it noticeably easier for organisations to promote opportunities and for volunteers to discover suitable activities, though the matching itself remained manual — the system surfaced opportunities but did not automatically pair volunteers with the most suitable tasks, and the evaluation sample was very small.

2.2 Preference-Based Volunteer Matching

Sudin, Mahrom, Thurairaju & Annuar (2026)

This study proposed a web-based volunteer scheduling system for sports events built around a Preference-Matching Algorithm that aligns volunteer roles with each individual's stated preferences, skills, and prior experience. Effectiveness was assessed using quantitative metrics such as match rate and average preference satisfaction. The results showed that preference-aware matching reduced event setup time, produced a fairer distribution of tasks, and increased overall volunteer satisfaction compared to manual scheduling. However, the approach is rule-based rather than adaptive: it was designed and validated specifically for sporting events, and the matching logic does not learn from outcomes over time.

2.3 NGO Support Software for Volunteer and Donation Management

Desale, Gautam, Khandare, Parikh & Toradmalle (2020)

The authors built a cross-platform NGO support application using Flutter and a Firebase backend, aimed at small NGOs that lack dedicated IT infrastructure. The system provides separate user and admin applications with modules for food donation, monetary donation (via an integrated payment gateway), volunteer registration, and social-media-style awareness posts, with all data stored and synchronised through Firebase in real time. Alpha and beta testing with small user groups (5 and 15 participants respectively) confirmed the system's usability and reasonable performance across devices. The platform is affordable and modular but relies entirely on manual administrative decisions — there is no AI-driven matching, no optimisation of resource allocation, and it is designed around a single NGO rather than a multi-organisation ecosystem.

2.4 Blockchain-Based Donation Tracking

Jadhav, Mohite & Pise (2025)

This research introduced Crypto-Go-Charity, a blockchain-based scholarship donation tracking platform involving three stakeholder roles: NGOs, donors, and scholarship beneficiaries. By recording every transaction on an immutable public ledger through smart contracts, the platform lets donors verify exactly how corporate social responsibility (CSR) funds are converted into student scholarships, addressing long-standing trust and accountability problems in the philanthropy sector. While the blockchain layer is effective at preventing fund misuse and building donor confidence, the system is narrowly scoped to financial tracking and does not extend to volunteer coordination, task allocation, or project-progress monitoring, and it assumes a level of technical/blockchain literacy that many everyday donors may not have.

2.5 Decentralised Charity Donation Platforms

Sahithi, Varsha, Dhanya & Nambiar P V (2025)

This work presented an Ethereum-based decentralised application for charity donations, implemented with Solidity smart contracts, the Hardhat development framework, and a MetaMask wallet for user transactions. The system creates a tamper-proof, end-to-end

auditable trail from donor to beneficiary, directly targeting the corruption and lack-of-accountability problems common in centralised donation systems. The trade-off is accessibility: donors must set up and manage a cryptocurrency wallet, transaction (gas) costs can vary, and the system, like other blockchain-based charity tools, focuses solely on the flow of funds rather than on volunteers, activities, or project outcomes.

2.6 GIS and Optimisation for Disaster-Relief Resource Allocation

Rodríguez-Espíndola, Albores & Brewster (2018)

This study combined Geographical Information Systems (GIS) with a bi-objective multi-organisational optimisation model to support disaster preparedness for floods. The GIS component analyses flood risk to identify safe locations for emergency facilities, while the optimisation model determines facility locations, relief-stock prepositioning, and the allocation of material and human resources across multiple collaborating organisations. Using real conditions from the 2013 floods in Acapulco, Mexico, the authors showed that a multi-organisational, GIS-driven approach reduces resource duplication and shortages compared to organisations acting independently. The model is powerful for one-off, large-scale crisis response, but it was not designed for the continuous, day-to-day coordination of volunteers and resources that routine NGO operations require.

2.7 Web-Based NGO Management and Donor-Engagement Platforms

Auti, Deshmukh, Sharma, Dhumal & Shinde (2023)

This paper proposed a web-based NGO management system with integrated modules for donor management, project tracking, volunteer coordination, event planning, and reporting, aiming to remove the communication bottlenecks and opaque information flow that typically exist between NGO leadership, staff, and beneficiaries. The system centralises administrative processes and gives stakeholders visibility into organisational progress, which the authors argue improves accountability and donor engagement. As with other general-purpose management platforms, the system's logic is essentially digitised record-keeping and dashboards rather than intelligent decision-making — it

Author / Year	Research Problem	Methodology / Approach	Dataset / Tools	Key Findings
Chen et al. (2011)	Difficulty recruiting suitable volunteers for non-profit activities	Mobile map-based matching prototype (bubble-icon interface)	Android prototype; interviews with 2 managers, 3 volunteers	Visual, location-aware browsing improved opportunity discovery and promotion
Sudin et al. (2026)	Manual scheduling causes unfair task allocation and long event setup time	Preference-Matching Algorithm (rule-based) for volunteer-role alignment	Web application; match-rate and preference-satisfaction metrics	Improved fairness, satisfaction, and reduced setup time for sports events
Desale et al. (2020)	Small NGOs lack affordable IT infrastructure for volunteers/donations	Cross-platform mobile app: food/money donation, volunteer & social modules	Flutter, Firebase Realtime DB, RazorPay; alpha/beta testing (5–15 users)	Low-cost, modular app improved reach, donation visibility, volunteer sign-up
Jadhav, Mohite & Pise (2025)	Lack of transparency/trust in scholarship and CSR donation tracking	Blockchain platform (Crypto-Go-Charity) with smart contracts	Public blockchain ledger, smart contracts; prototype design	End-to-end fund traceability improved donor trust and accountability
Sahithi et al. (2025)	Centralised donation systems are prone to fraud and poor accountability	Ethereum dApp using Solidity smart contracts	Hardhat framework, MetaMask wallet, Ethereum testnet	Tamper-proof, fully auditable donor-to-recipient donation trail
Rodríguez-Espíndola et al. (2018)	Multi-agency disaster response requires optimal deployment of scarce resources	GIS-based risk mapping combined with bi-objective optimisation (MIP)	GIS software, optimisation model; 2013 Acapulco flood case study	More efficient organisation/resource combinations; reduced duplication
Auti et al. (2023)	Opaque communication and progress tracking between NGOs and stakeholders	Web-based platform: donor management, project tracking, volunteer modules	Web application (design and implementation study)	Streamlined workflows; improved transparency and stakeholder communication

5. Critical Analysis and Identified Limitations

Reviewing the seven studies together reveals a consistent pattern: each addresses one functional slice of the problem statement well, but none integrates all of them into a single coherent platform.

- Volunteer-matching research (Chen et al., 2011; Sudin et al., 2026) shows that structured, preference- or map-based matching genuinely improves fairness, satisfaction, and coordination speed, but existing systems are either purely browsing tools with no automated matching, or rule-based algorithms tuned to one specific domain (e.g., sports events) rather than a general, adaptive, AI-driven recommender that learns from volunteer history and feedback.
- Blockchain-based donation-tracking research (Jadhav et al., 2025; Sahithi et al., 2025) convincingly demonstrates that a tamper-proof ledger builds donor trust and prevents fund misuse, but every reviewed system treats donation tracking as an isolated financial module — none links the ledger to volunteer activity, project milestones, or resource needs, and all impose a technical/crypto-literacy barrier on ordinary donors.
- General-purpose NGO management platforms (Desale et al., 2020; Auti et al., 2023) succeed in digitising day-to-day administrative work and improving stakeholder communication, but their 'intelligence' stops at dashboards and manual record-keeping; they contain no predictive or optimisation capability to anticipate resource shortfalls or recommend better volunteer-task pairings.
- Disaster-relief optimisation research (Rodríguez-Espíndola et al., 2018) proves that GIS and mathematical optimisation can meaningfully improve multi-agency resource allocation, but this class of model is built for episodic, large-scale crisis response rather than the continuous, everyday resource-matching problem that routine NGO operations face.

6. Research Gap

Across the reviewed literature, no single system combines (a) AI-based, adaptive volunteer-to-task matching that learns from outcomes, (b) transparent and accessible donation/resource tracking that does not require specialised blockchain literacy, (c)

GIS- or optimisation-informed allocation of volunteers and resources for everyday (not just disaster-scale) operations, and (d) real-time, predictive project-monitoring dashboards for administrators — all within one unified, multi-NGO platform. Existing research instead solves these problems in isolation, leaving an open gap for an integrated AI-powered smart volunteer and resource management platform of the kind described in the assigned problem statement.

7. Future Directions

- Develop adaptive, machine-learning-based volunteer-matching models that build on the preference-matching approach of Sudin et al. (2026) but learn continuously from volunteer feedback, performance history, and evolving project needs, rather than relying on a fixed rule set.
- Layer a lightweight, donor-friendly transparency ledger over conventional NGO management systems, preserving the accountability benefits demonstrated by Jadhav et al. (2025) and Sahithi et al. (2025) while removing the cryptocurrency-wallet barrier for everyday donors.
- Extend GIS- and optimisation-based allocation techniques from disaster logistics (Rodríguez-Espíndola et al., 2018) to routine, everyday NGO operations — for example, matching volunteer location and availability to the nearest active project need.
- Embed predictive analytics into project-monitoring dashboards, building on the descriptive dashboards of Auti et al. (2023) and Desale et al. (2020), so administrators are alerted to at-risk projects or impending resource shortages before they occur.
- Design the platform for multi-NGO, federated use rather than a single organisation, enabling resource- and volunteer-sharing across collaborating NGOs, echoing the multi-organisational coordination principle demonstrated in disaster-relief research.

8. Conclusion

This literature survey reviewed seven research contributions spanning AI- and preference-based volunteer matching, blockchain-based donation transparency, GIS-

driven disaster resource optimisation, and general-purpose NGO management platforms. Each stream of work makes a genuine contribution to one aspect of the problem — matching, transparency, optimisation, or administration — yet none combines all of them into a single integrated system. The comparative analysis confirms a clear research gap: an AI-powered platform that unifies intelligent volunteer matching, transparent and accessible donation/resource tracking, optimisation-based resource allocation, and real-time predictive project monitoring for NGOs. Addressing this gap, as targeted by the assigned problem statement, would directly support the expected outcomes of improved volunteer participation, efficient resource allocation, enhanced transparency, greater donor confidence, and ultimately, greater social impact.

9. References

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